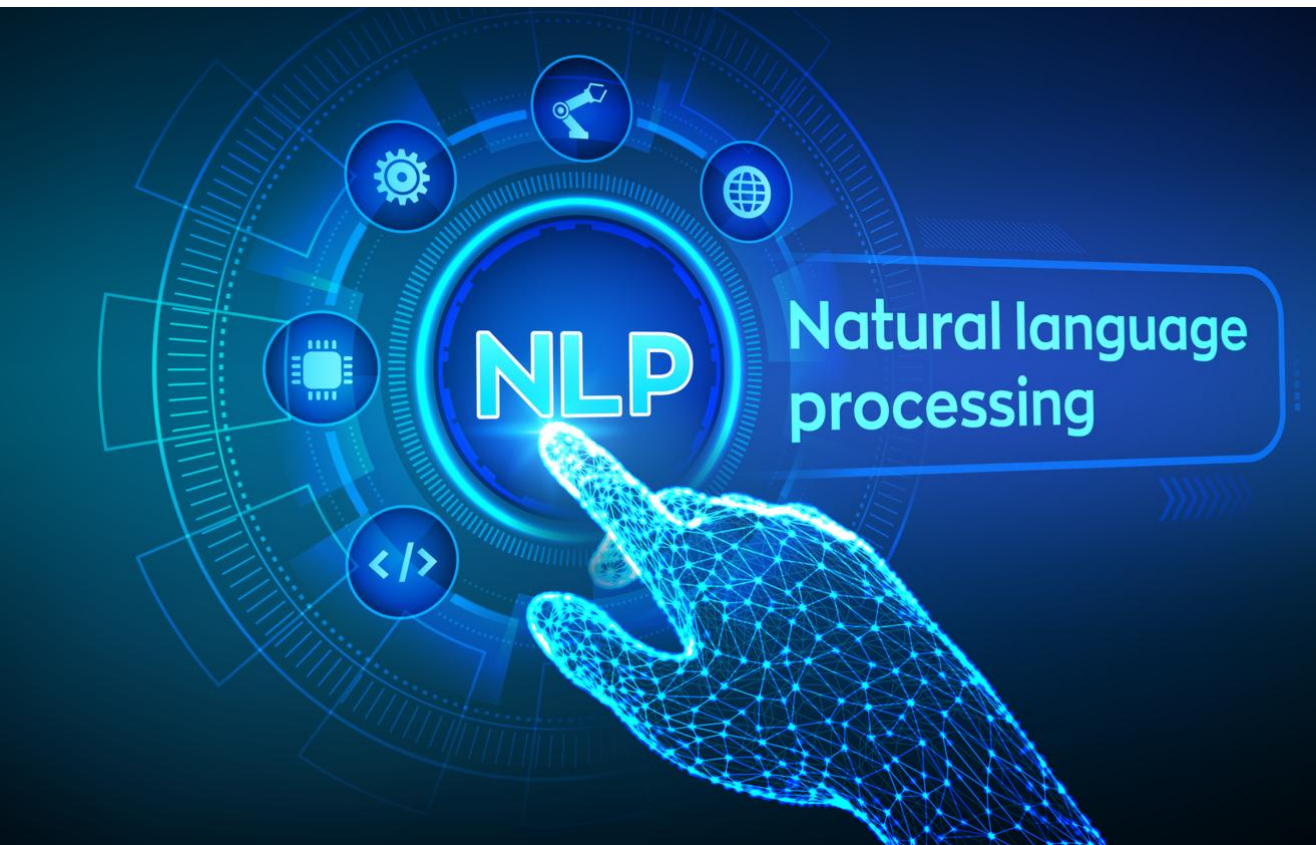




MORPHOLOGY AND FINITE-STATE TRANSDUCERS

27.01.2026

Unit II

Morphology and Finite-State Transducers: Inflectional Morphology - Derivational Morphology- Finite-State Morphological Parsing- The Lexicon and Morphotactics - Morphological Parsing with Finite-State Transducers- Combining FST Lexicon and Rules- Lexicon-free FSTs: The Porter Stemmer- Human Morphological Processing- Speech Sounds and Phonetic Transcription- The Phoneme and Phonological Rules.

Content

-
- ✓ Morphology And Finite-State Transducers
 - ✓ Inflectional Morphology
 - ✓ Derivational Morphology



Morphology

Morphology

- Morphology is the study of the way words are built up from smaller meaning bearing units, morphemes.
- A morpheme is often defined as the minimal meaning-bearing unit in a language.
- For example the word **fox** consists of a single morpheme (the morpheme fox) while the word **cats** consists of two: the morpheme **cat** and the morpheme **-s**

Types of Morpheme

- One Morpheme — Nation
- Two Morpheme — National (nation, al)
- Three Morpheme — Nationalize (nation, al, ize)
- Four Morpheme — Denationalize (de, nation, al, ize)

All morphemes

Variants of same morpheme

- Assert/ **Assertion**
- Electric/ **Electricity**
- Impress/ **Impression**
- Include/ **Inclusive**

Bounded morphemes

When it is added to a morpheme it gives meaning

- **s** in walks
- **re** in replay
- **er** in cheaper
- **im** in impossible
- **en** in enlighten
- **un** in unable

Free morphemes

- It stand alone as a word
- Eg: Girl, cat, dog, little, Book, Bag

Stem and Affixes

- Two broad classes of morphemes, **stems and affixes**:
 - ✓ **Stem** is the ‘main’ morpheme of the word, supplying the main meaning,
 - ✓ **Affixes** add ‘additional’ meanings of various kinds

Contd...

- Affixes are further divided into prefixes, suffixes, infixes, and circumfixes.
 - ✓ **Prefixes** precede the stem
 - ✓ **Suffixes** follow the stem
 - ✓ **Circumfixes** do both
 - ✓ **Infixes** are inserted inside the stem

Examples

- **Prefixes:** **im**possible, **re**ply, **un**happy, **con**firm, **com**press
- **Suffixes:** Pass**ion**, Ambit**ion**, Unit**y**, Walk**ing**
- **Circumfixes:** **en**light**en**, **em**bold**en**

Example

- ✓ The word **eats** is composed of a stem **eat** and the suffix **-s**.
- ✓ The word **unbuckle** is composed of a stem **buckle** and the prefix **un-**.

Concatenative Morphology

- ✓ **Prefixes and suffixes** are often called **concatenative morphology** since a word is composed of a number of morphemes concatenated together.
- ✓ Circumfixes (Not in English)
Eg. In German, for example
 - ✓ The past participle of some verbs formed by adding ge- to the beginning of the stem and -t to the end;
 - ✓ so the past participle of the verb sagen (to say) is gesagt (said).

Non-Concatenative Morphology

- ✓ A number of languages have extensive non-concatenative morphology, in which morphemes are combined in **more complex ways**
- ✓ Another kind of non-concatenative morphology is called **templatic morphology or root and-pattern morphology**. This is very common in Arabic, Hebrew, and other Semitic languages.

Contd...

- ✓ In Hebrew, for example, a verb is constructed using two components: a root, consisting usually of three consonants (CCC) and carrying the basic meaning, and a template, which gives the ordering of consonants and vowels and specifies more semantic information about the resulting verb, such as the semantic voice (e.g. active, passive, middle).

Example

- ✓ The Hebrew tri-consonantal root lmd, meaning 'learn' or 'study', can be combined with the active voice CaCaC template to produce the word lamad, 'he studied'.
- ✓ The intensive CiCeC template to produce the word limed, 'he taught'.
- ✓ The intensive passive template CuCaC to produce the word lumad, 'he was taught'.

Two broad classes of ways to form words from morphemes



INFLECTION



DERIVATION

Inflection

- Inflection is the combination of a word stem with a grammatical morpheme, usually resulting in a word of the same class as the original stem, and usually filling some **syntactic function**, e.g. plural of nouns.
- *For example, English has the inflectional morpheme -s for marking the plural on nouns, and the inflectional morpheme -ed for marking the past tense on verbs*
- Inflection is **productive**.
- The meaning of the resulting word is easily **predictable**.

Derivation

- Derivation is the combination of a word stem with a grammatical morpheme, usually resulting in a word of a different class, often with a **meaning hard to predict exactly**.
- For example the verb **computerize** can take the derivational suffix **-ation** to produce the noun **computerization**.
- Not always productive.



Inflection morphology

Inflection morphology

- English has a relatively simple inflectional system; only nouns, verbs, and sometimes adjectives can be inflected, and the number of possible inflectional affixes is quite small.
- English nouns have only two kinds of inflection:
 - ✓ an affix that marks plural
 - ✓ an affix that marks possessive

Contd...

- For example, many English nouns can either appear in the bare stem or singular form, or take a plural suffix.
- Examples of the regular plural suffix -s,
- the alternative spelling -es, and irregular plurals:

	Regular Nouns		Irregular Nouns	
Singular	cat	thrush	mouse	ox
Plural	cats	thrushes	mice	oxen

Contd...

- While the regular plural is spelled -s after most nouns,
- It is spelled -es after words ending in
 - ✓ -s (ibis/ibises)
 - ✓ -z, (waltz/waltzes)
 - ✓ -sh, (thrush/thruses)
 - ✓ -ch, (finch/finches)
 - ✓ -x (box/boxes)

Contd...

- Nouns ending in -y preceded by a consonant change the -y to -i (butterfly/butterflies).
- The possessive suffix is realized by apostrophe +
 - ✓ -s for regular singular nouns (llama's)
 - ✓ Plural nouns not ending in -s (children's)
 - ✓ Often by a lone apostrophe after regular plural nouns (llamas')
 - ✓ Some names ending in -s or -z (Euripides' comedies)

Contd...

- English verbal inflection is more complicated than nominal inflection. First, English has three kinds of verbs;
 - ✓ Main verbs (eat, sleep, impeach)
 - ✓ Modal verbs (can, will, should)
 - ✓ Primary verbs (be, have, do)

Contd...

- These regular verbs (e.g. walk, or inspect), have four morphological forms

Morphological Form Classes	Regularly Inflected Verbs			
stem	walk	merge	try	map
-s form	walks	merges	tries	maps
-ing participle	walking	merging	trying	mapping
Past form or -ed participle	walked	merged	tried	mapped

- These verbs are called regular because just by knowing the stem we can predict the other forms, by adding one of three predictable endings, and making some regular spelling changes.

Contd...

- The irregular verbs are those that have some more or less idiosyncratic forms of inflection.
- Irregular verbs in English often have five different forms, but can have as many as eight (e.g. the verb be) or as few as three (e.g. cut or hit).
- While constituting a much smaller class of verbs estimate there are only about 250 irregular verbs, not counting auxiliaries), this class includes most of the very frequent verbs of the language.

Sample irregular forms

Morphological Form Classes	Irregularly Inflected Verbs		
stem	eat	catch	cut
-s form	eats	catches	cuts
-ing participle	eating	catching	cutting
Past form	ate	caught	cut
-ed participle	eaten	caught	cut

- An irregular verb can inflect in the past form (also called the preterite) by changing its vowel (eat/ate), or its vowel and some consonants (catch/caught), or with no ending at all (cut/cut).

Contd...

- The **-s form** is used in the 'habitual present' form to distinguish the **3rd-person singular ending** (She jogs every Tuesday) from the other choices of person and number (I/you/we/they jog every Tuesday).
- The stem form is used in the infinitive form, and also after certain other verbs (I'd rather walk home, I want to walk home)
- The -ing participle is used when the **verb** is treated as a **noun** - called a gerund use.

eg. Fishing is fine if you live near water

Contd...

- The -ed participle is used in
 - ✓ Perfect construction (He's eaten lunch already) or
 - ✓ Passive construction (The verdict was overturned yesterday.)

Contd...

- A **single consonant letter** is doubled before adding the -ing and -ed suffixes (beg/begging/begged).
- If the **final letter is 'c'**, the doubling is spelled 'ck' (picnic/picnicking/picnicked).
- If the **base ends in a silent -e**, it is deleted before adding -ing and -ed (merge/merging/merged).
- Just as for nouns, the **-s ending is spelled -es** after verb stems ending in -s (toss/tosses) , -z, (waltz/waltzes) -sh, (wash/washes) -ch, (catch/catches) and sometimes -x (tax/taxes).

Contd...

- Also like nouns, verbs ending in -y preceded by a consonant change the -y to -i (try/tries).
- The English verbal system is much simpler than for example the European Spanish system, **fifty distinct verb** forms for **each regular verb**.



Derivational Morphology

Derivation Morphology

- Derivation in English is quite complex. It is the combination of a word stem with a grammatical morpheme, usually resulting in a word of a different class, often with a meaning hard to predict exactly.
- A very common kind of derivation in English is the formation of new nouns, often from verbs or adjectives. This process is called **nominalization**.
- For example, the suffix -ation produces nouns from verbs ending often in the suffix -ize (computerize ! computerization).

Examples- English nominalizing suffixes

Suffix	Base Verb/Adjective	Derived Noun
-ation	computerize (V)	computerization
-ee	appoint (V)	appointee
-er	kill (V)	killer
-ness	fuzzy (A)	fuzziness

Adjectives can also be derived from nouns and verbs - Examples

Suffix	Base Noun/Verb	Derived Adjective
-al	computation (N)	computational
-able	embrace (V)	embraceable
-less	clue (N)	clueless

Contd...

- Derivation in English is more complex than inflection for a number of reasons.
 1. It is generally less productive; even a nominalizing suffix like **-ation**, which can be added to almost any verb ending in **-ize**, cannot be added to absolutely every verb.
- Thus we can't say ***eatation** or ***spellation** (asterisk (*) to mark 'non-examples' of English).
- 2. There are subtle and complex meaning differences among nominalizing suffixes.
- For example **sincerity** has a subtle difference in meaning from **sincereness**.



**THANK
YOU!**